



**When the signatures
break, does your
identity break with
them?**

The Post-Quantum Identity Blueprint.

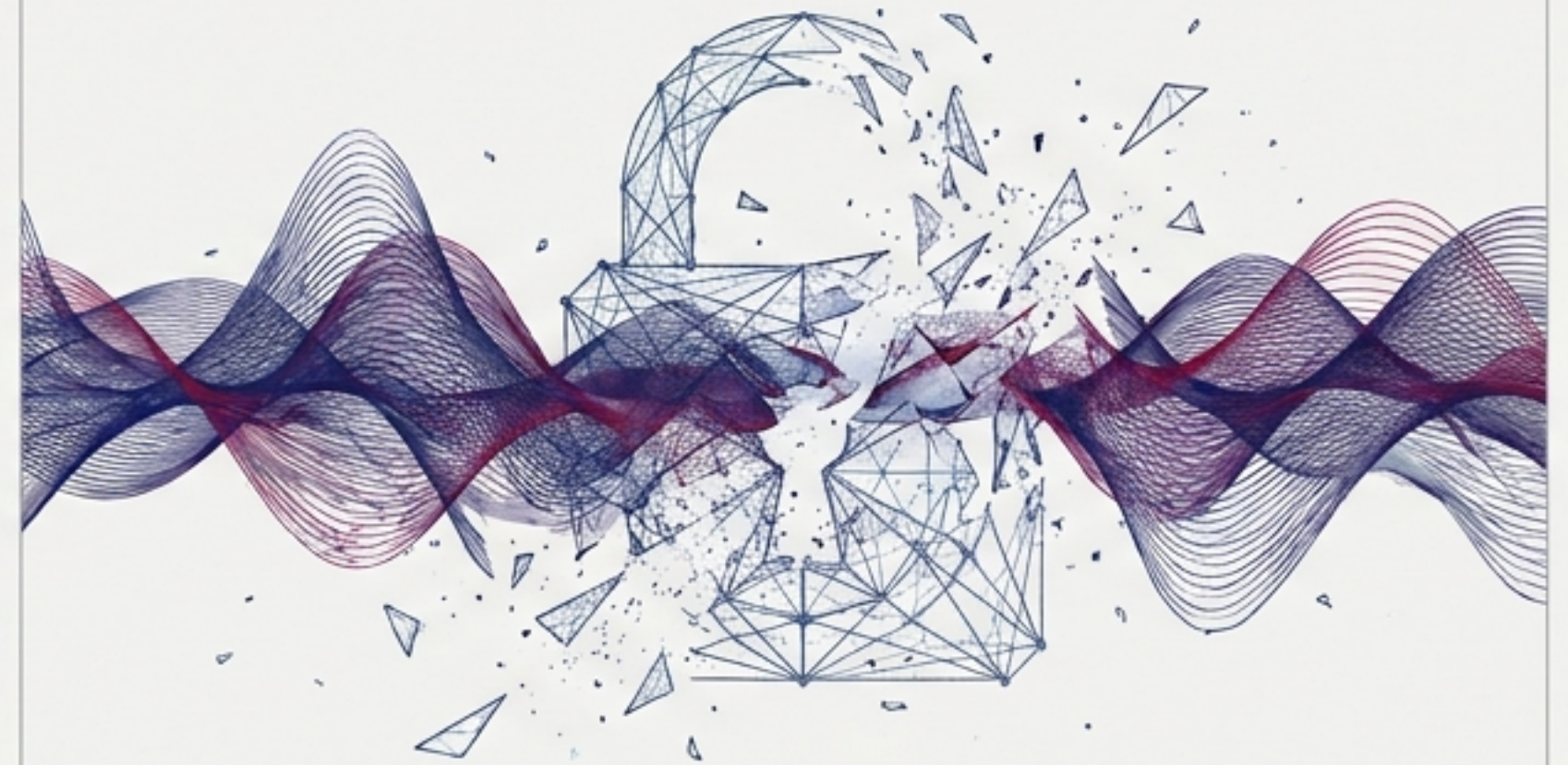
Navigating the migration of self-sovereign identity infrastructure
and KERI witnesses in a post-quantum world.

The Vulnerability of a Signed History

A digital identity is fundamentally a running, signed history. Those signatures rest on mathematics that a quantum computer will eventually be able to reverse.



Today's Signature (Ed25519):
Proof an entry came from you.

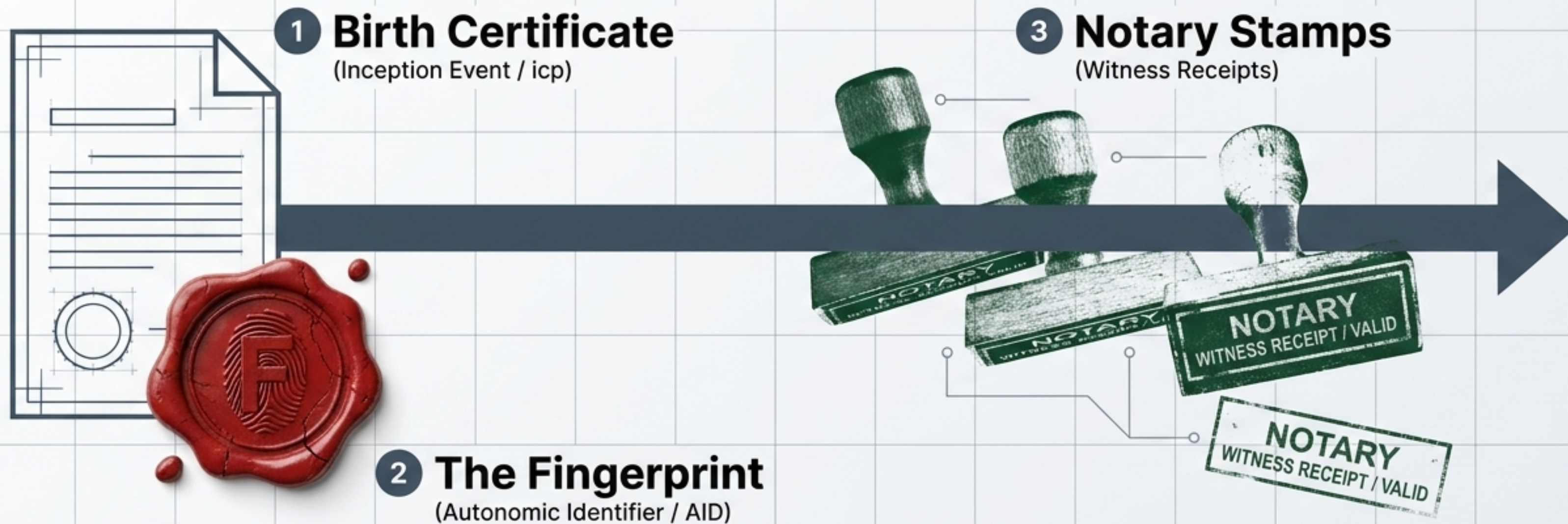


The Quantum Threat: A computer powerful
enough to forge that signature.

If a forger can sign as me, can't they rewrite my history or simply drop my safeguards?

Anatomy of a Self-Sovereign Identity

Every action in your digital identity is mathematically chained back to its origin.
To break the identity, an attacker must break this chain.

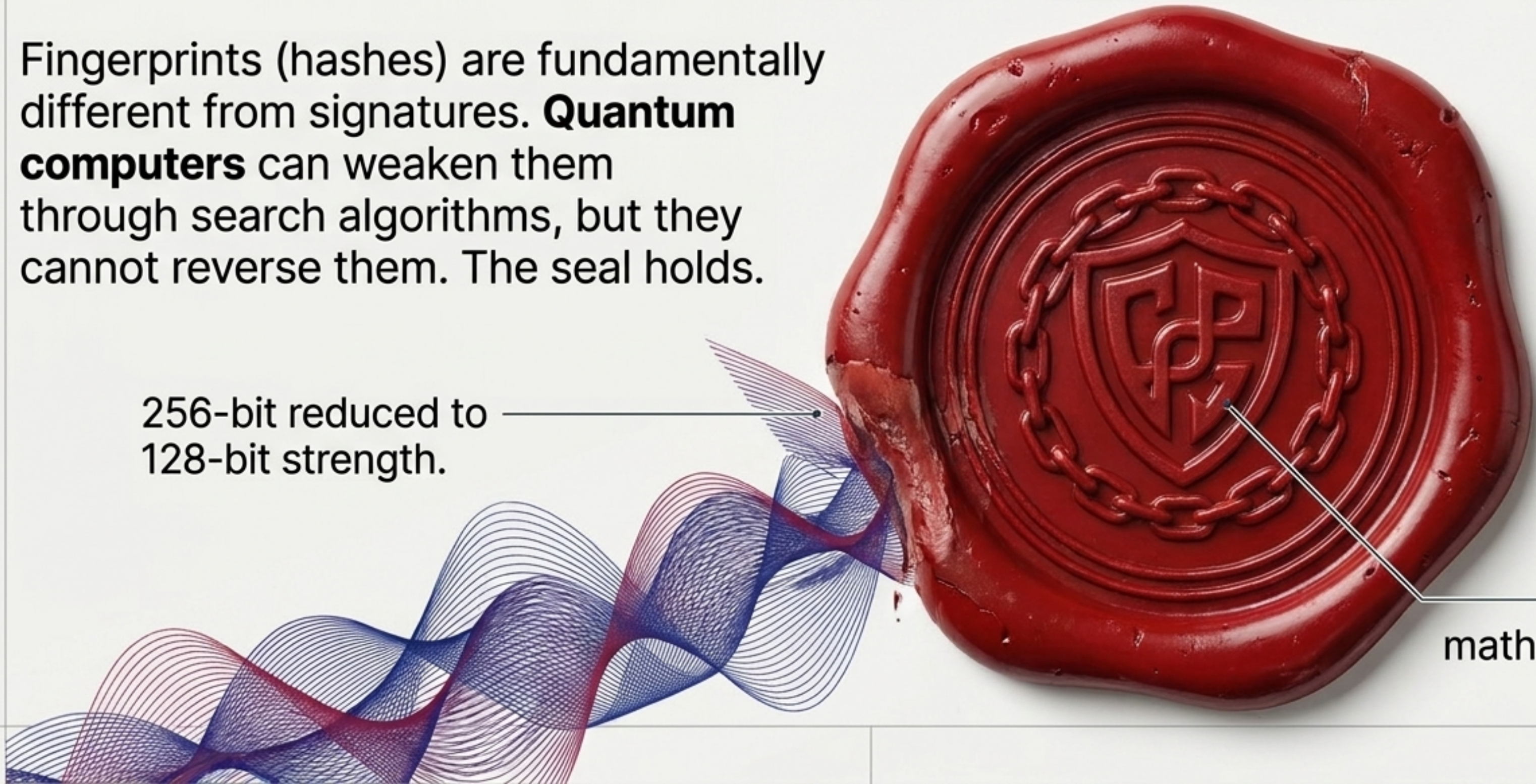


Shield One: They Cannot Change Your Rules

Your identity's name is a direct fingerprint of its founding record.

Fingerprints (hashes) are fundamentally different from signatures. **Quantum computers** can weaken them through search algorithms, but they cannot reverse them. The seal holds.

256-bit reduced to
128-bit strength.



Still
mathematically
secure.

The Identity Trap

The rules of your identity are locked into the founding fingerprint.

The Impersonation Attempt

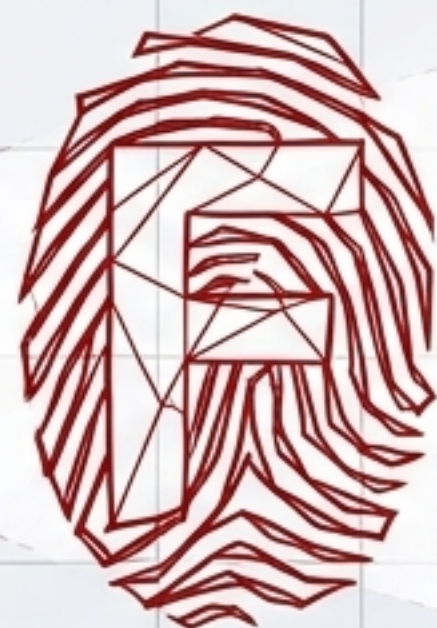
1. Original Record



2. Attacker erases safeguards



3. Recalculated Fingerprint



If an attacker tries to create a safeguard-stripped version of your history, the fingerprint mathematically changes. It becomes a brand-new identity that **impersonates no one**.

Shield Two: Notaries Stamp What They See First

Witnesses act as independent notaries. They record ordinary actions (interactions) and refuse to endorse a conflicting version of events.



A forged fork of your history can never gather enough stamps to be believed by the network.

The Two Shields: What Protects What?

Pre-rotation stops an attacker from seizing lasting control in the future. But it cannot stop a forged ordinary action today. That is exactly what Witnesses are built for.

	Locks the Future (Key Changes)	Guards the History (Ordinary Actions)
Pre-Rotation	 Seals next keys behind a fingerprint	 Does not stop forged interactions
Witnesses	 Does not stop forged interactions	 Detects and rejects conflicting timelines

The Hidden Vulnerability

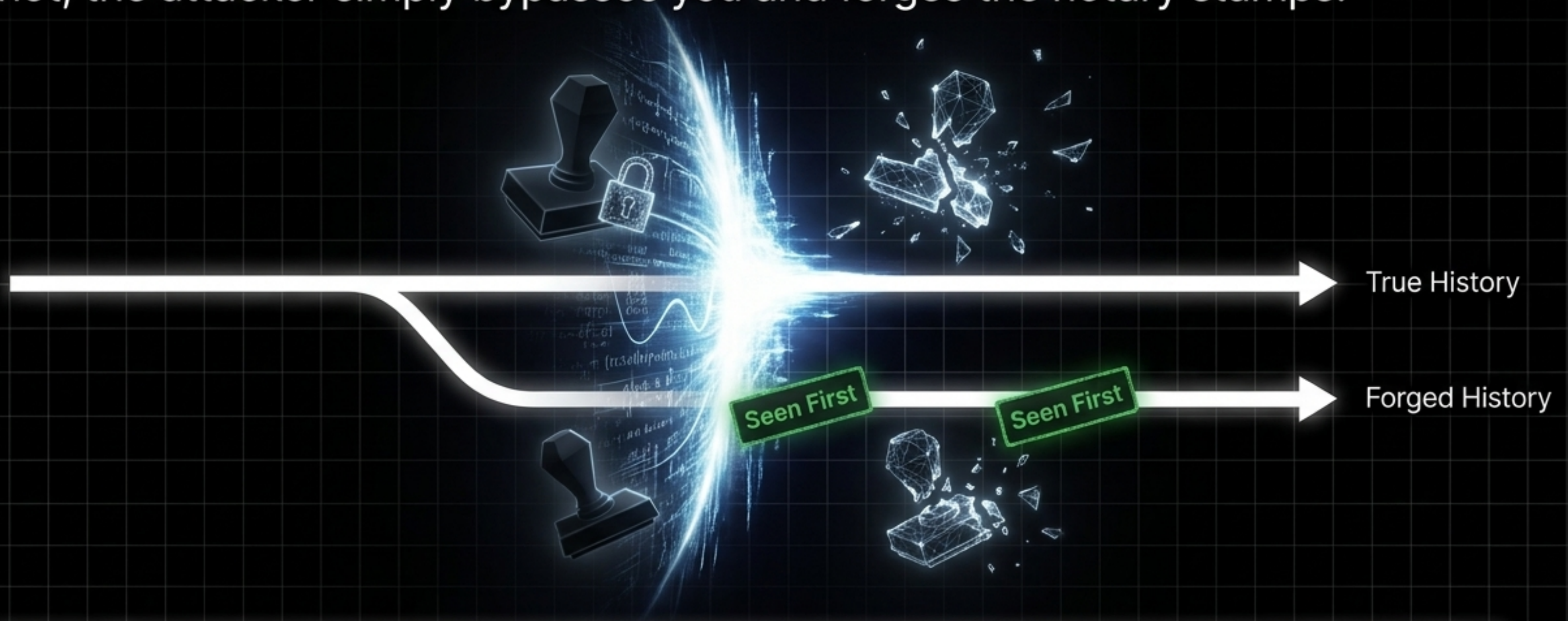
The safeguard is made of the same breakable mathematics.

Witness stamps are themselves signed. Usually, with the exact same legacy cryptography that the quantum computer is designed to break.



Headline: The Forgery Scenario

Main Concept: If your identity upgrades to quantum-safe keys, but your witnesses do not, the attacker simply bypasses you and forges the notary stamps.



Protecting the identity is meaningless if the safeguards guarding it can be counterfeited.

Bounding the Threat: A Safe Denial

Even in a worst-case scenario where witness stamps are forged, the attacker never gets to wear your identity. The system defaults to a safe denial, freezing the identity rather than handing over control.



Cannot Seize the Future

Pre-rotation securely locks the rotation path.



Cannot Fake the Origin

The identity fingerprint is permanent and unforgeable.



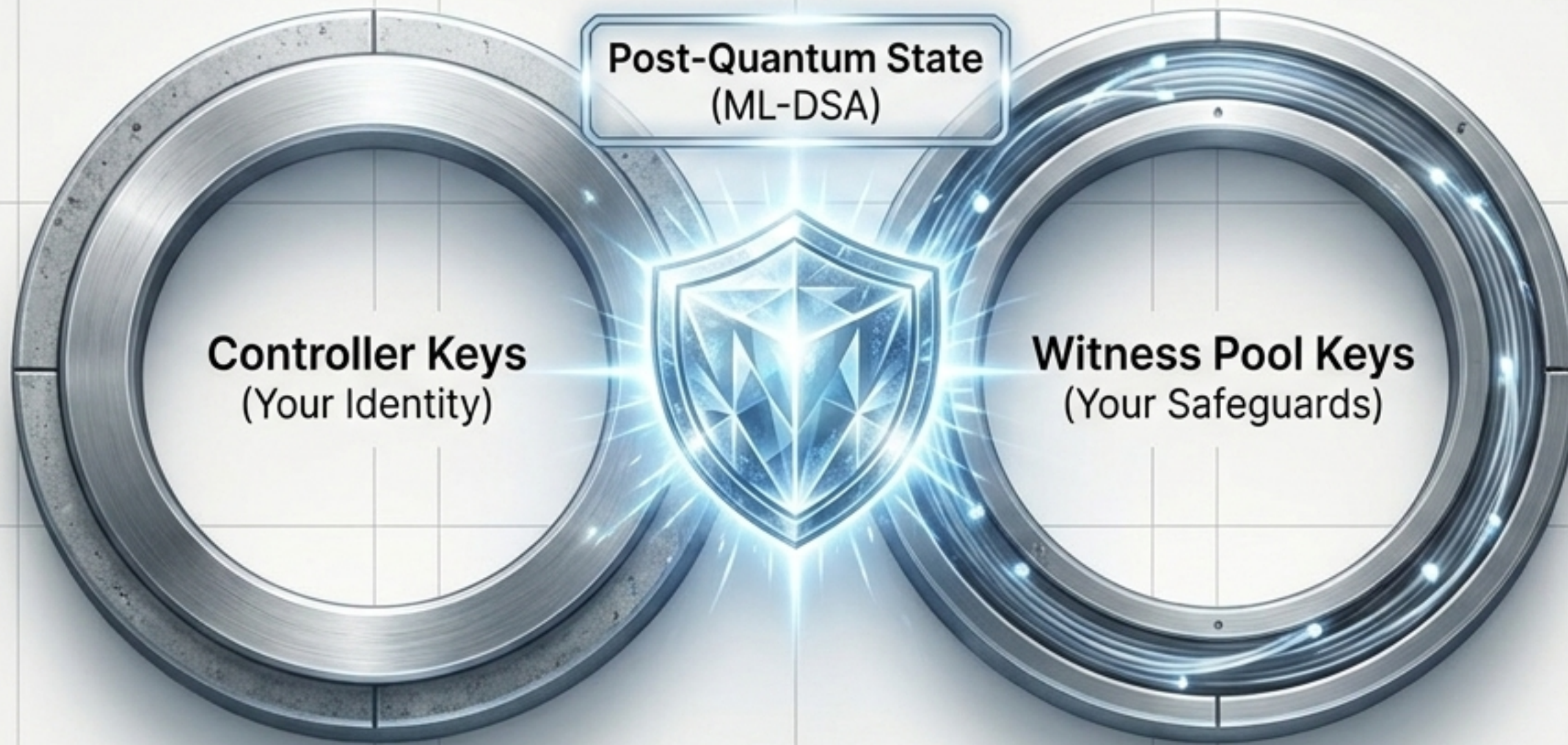
The Worst Case is a Freeze

If witness stamps are forged, verifiers see a conflict and halt all trust.

The Mandate: One Migration, Not Two

Upgrading your own keys is not enough. To survive the quantum transition, you must rotate controller keys and witness keys simultaneously.

Everything else in the self-sovereign architecture is already built to outlive legacy signatures. This **dual-rotation** is the final requirement.



Post-Quantum Identity Checklist



Treat AIDs as Permanent:

Fingerprints withstand quantum pressure. Never attempt to strip witness requirements.



Rely on Witnesses for Interactions: Understand that pre-rotation secures the future, but witnesses secure the daily history.



Mandate Simultaneous Migration: Treat controller key upgrades and witness pool upgrades as a single, indivisible operation.



Trust the Failsafe: Communicate to stakeholders that a post-quantum cryptographic failure results in an operational freeze, not an identity breach.